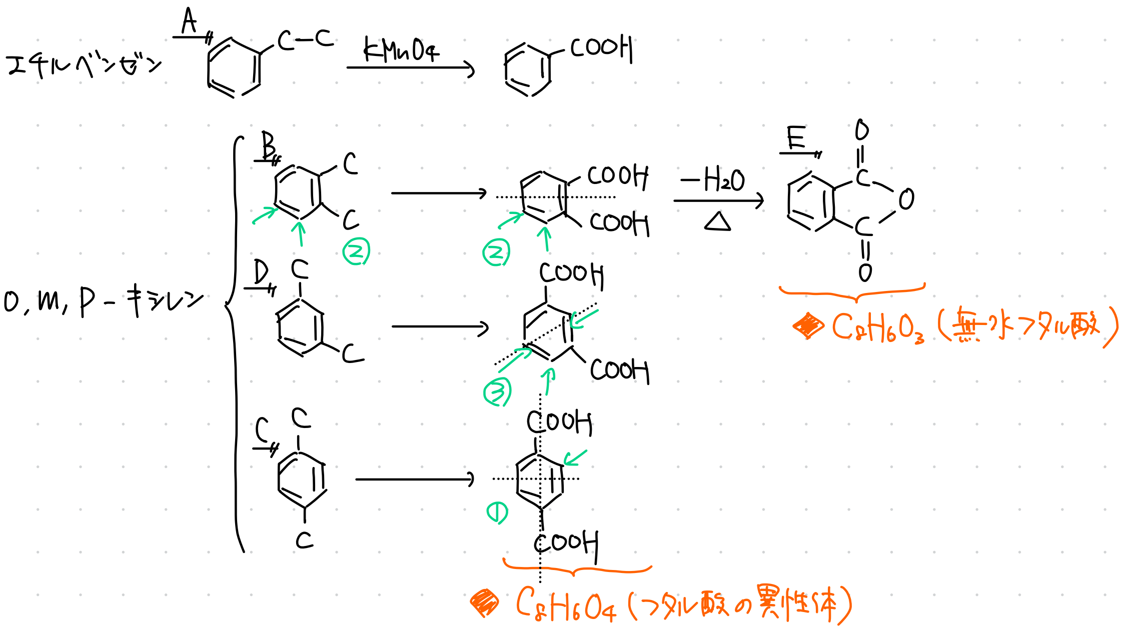
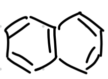
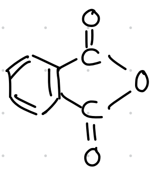
4-[1]　C_8H_{10}　$U=4$  ×1



問2　Ox　$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$　) ×9

Red　$C_{10}H_8 + 7H_2O \longrightarrow C_8H_4O_3 + 2CO_2 + 18H^+ + 18e^-$　) ×2

---

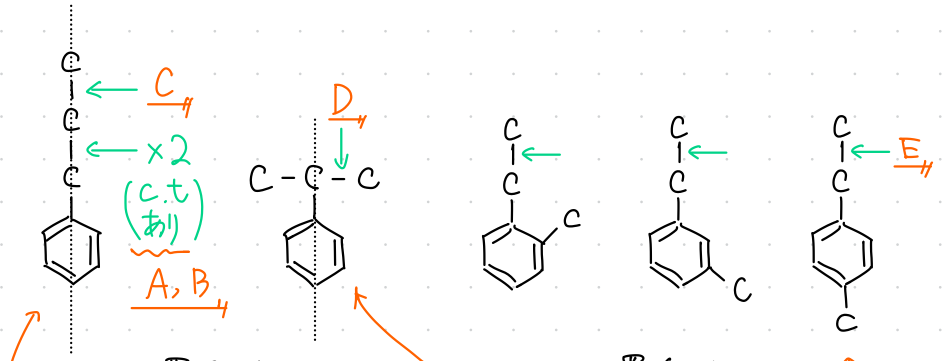
2  $+ 9O_2 \longrightarrow$ 2  $+ 4CO_2 + 4H_2O$ //

(酸化だから，余ったC $\xrightarrow{O_2}$ CO_2,
H $\xrightarrow{O_2}$ H_2O になると考えて，
係数合わせても良い)

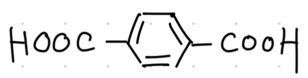
4-②　C_9H_{10}　$U=5$

 ×1 + C=C ×1
(U=4)　(U×1)

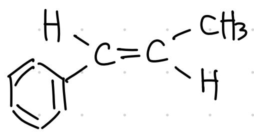
問2.

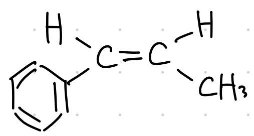


<1置換体>　　<2置換体>

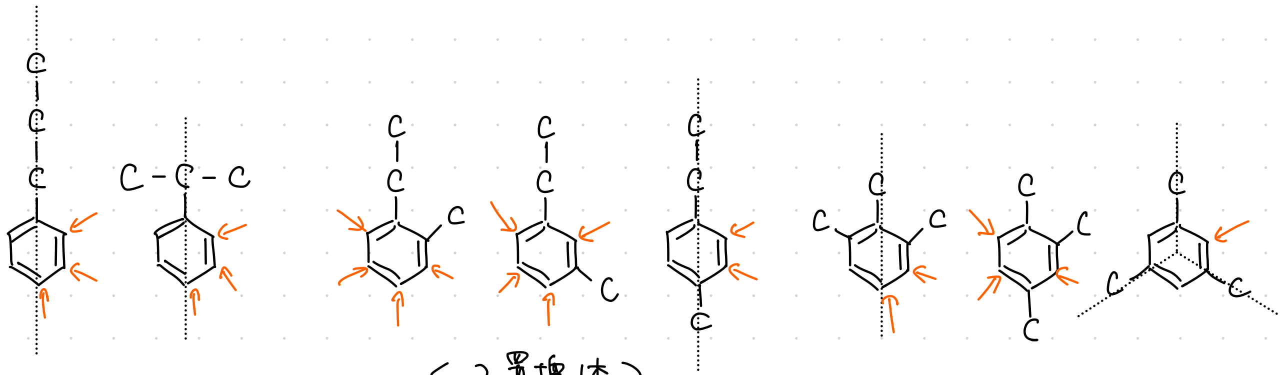
$$\text{A, B, C} \xrightarrow{H_2} \text{F}$$
$$\text{D} \xrightarrow{H_2} \text{G : 7メン}$$
$$\text{E} \xrightarrow{H_2} \text{H} \xrightarrow[O_2]{Ox}$$ 

A, B : シス・トランス異性体
融点 A > B
トランス　シス

A : 

B : 

問1. 8　　問3. 22



<1置換体>
Cl置換 : 6

<2置換体>
Cl置換 : 10

<3置換体>
Cl置換 : 6

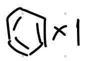
4−3

問1　$C_lH_mO_n + \underline{\quad\quad} O_2 \longrightarrow l\,CO_2 + \frac{m}{2}H_2O$

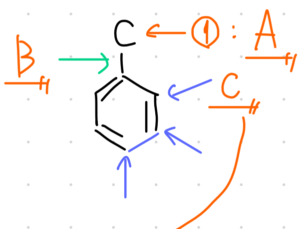
条件より　$l : \frac{m}{2} = 7 : 4$　∴ $(l, m) = (7, 8)$

∴ $(C_7H_8O_n)_k = (92 + 16n)k = 108$

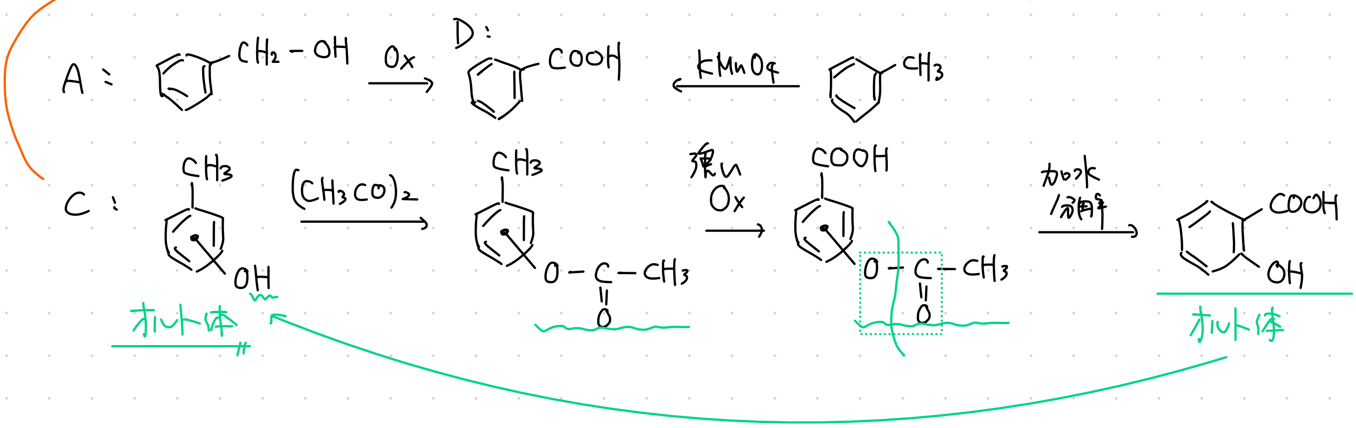
∴ $(n, k) = (1, 1)$　　$\underline{C_7H_8O}$

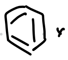
問2.　C_7H_8O　$U = 4$ 

{ アルコール / フェノール類 / エーテル

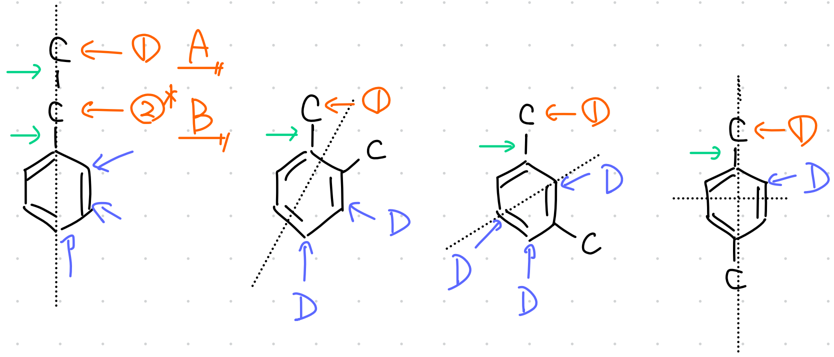


	A	B (エーテル)	C
Na	$H_2\uparrow$	×	
NaOH	× (アルコール)		とける (フェノール類)



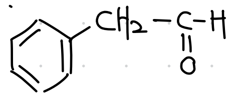
4-$\boxed{4}$　$C_8H_{10}O$　$U=4$  ×1

- アルコール
- フェノール類
- エーテル

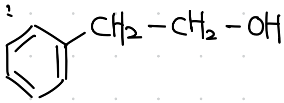


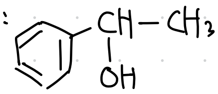
	第1級アルコール A	B	C	D
置換基	1, 2			3
おだやか Ox	E アルデヒド			
脱水	F	F		
$FeCl_3$ aq			−	+

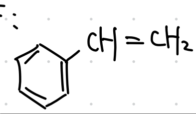
AとBは同一C骨格

E: 

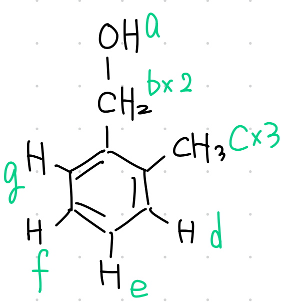
↑ おだやか酸化

A: 

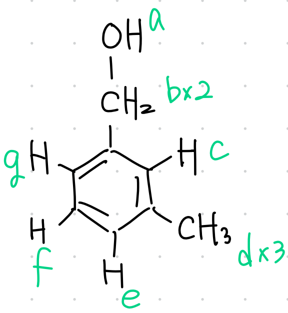
B: 

→ F: 

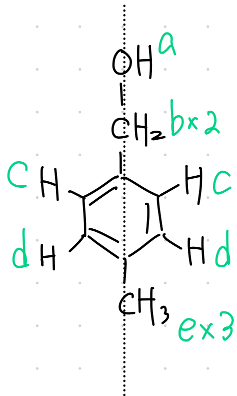
C : 1or 2置換体のアルコール



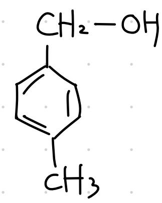
7種 X



7種 X



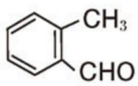
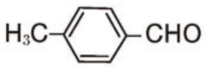
5種 (1:2:2:2:3)

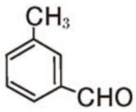
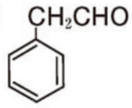
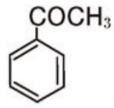
= C: 

問2

D: フェノール類（3置換体）, 6

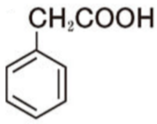
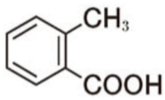
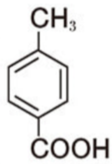
4－$\boxed{5}$

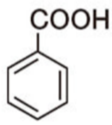
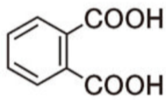
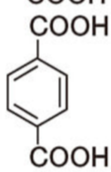
(1) A  B 

C  D  E 

(2) **G は *p*-置換体であるため，*o*-，*m*-置換体よりも分子の形が対称的である。したがって，結晶化しやすいため，融点は最も高くなる。**

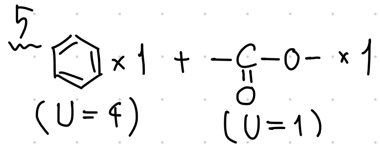
4－$\boxed{6}$

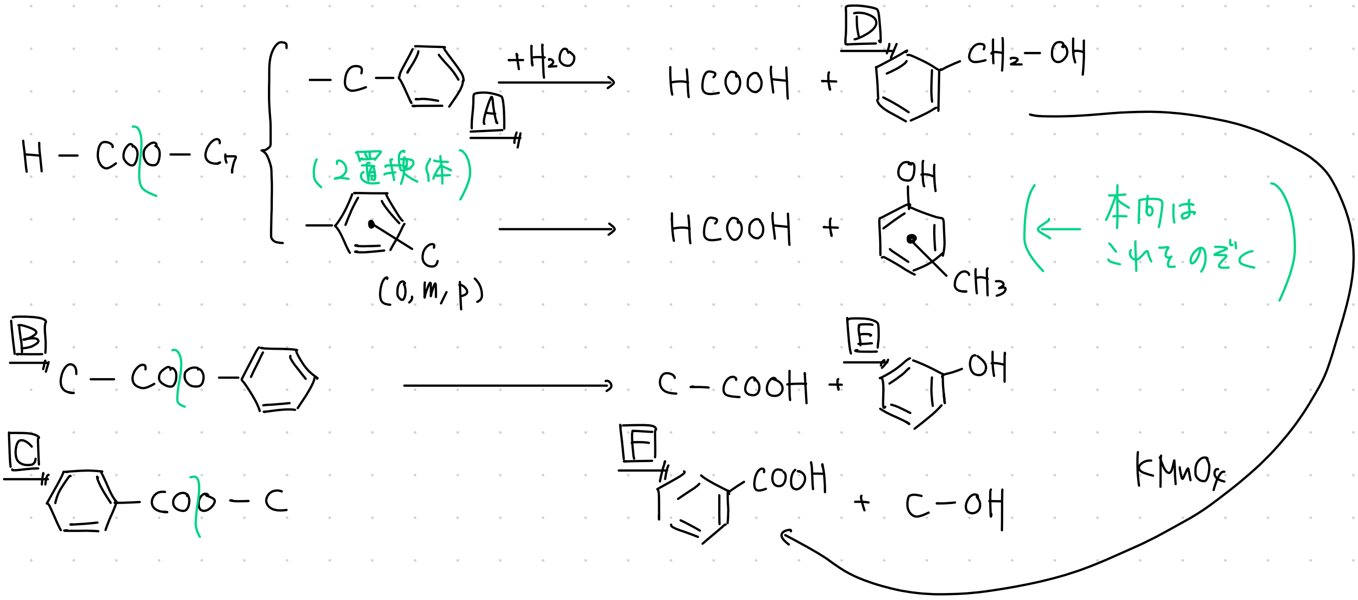
A  B  C 

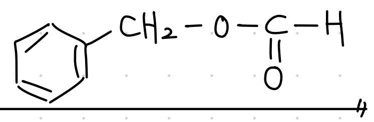
D  E  F 

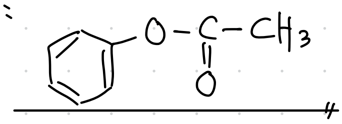
4-$\boxed{7}$

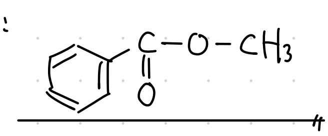
問1 $C_8H_8O_2$ U=5



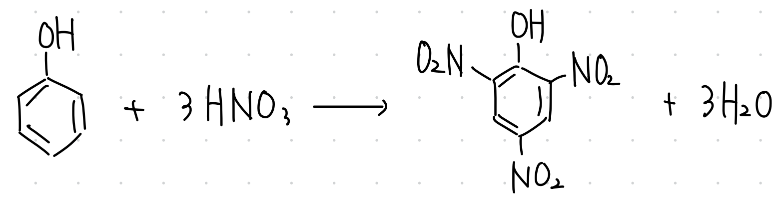


A:


B:


C:


問2. E

問3


ピクリン酸

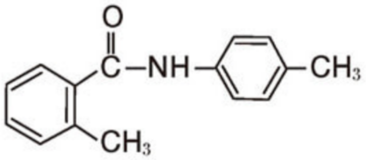
(2,4,6 − トリニトロフェノール)

4 − $\boxed{8}$

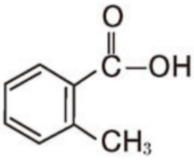
(1) $C_{15}H_{15}NO$　　$U = \dfrac{2\times15+2+1-15}{2} = 9$

(2)

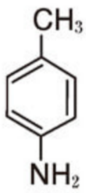
A



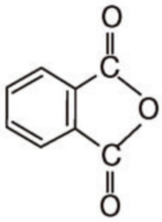
B



C



D



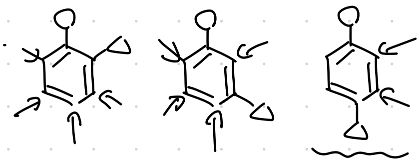
$C_{15}H_{15}NO$　　$U = \dfrac{2\times15+2+1-15}{2} = 9 = \underset{\text{アミド}}{-\mathrm{C}(=\mathrm{O})-\mathrm{NH}-}$ +  ×2

アミドA $\xrightarrow[\mathrm{H^+}]{\text{加水分解}}$ (エ) カルボン酸 B (オルト体) $\xrightarrow{\mathrm{KMnO_4}}$ B′ (オルト体) $\xrightarrow[\Delta]{-\mathrm{H_2O}}$ 酸無水物D

C₈以上

→ (水) アミンCのHCl塩 $\xrightarrow[\text{WBゆーり}]{\mathrm{OH^-}}$ アミンC

・2置換体

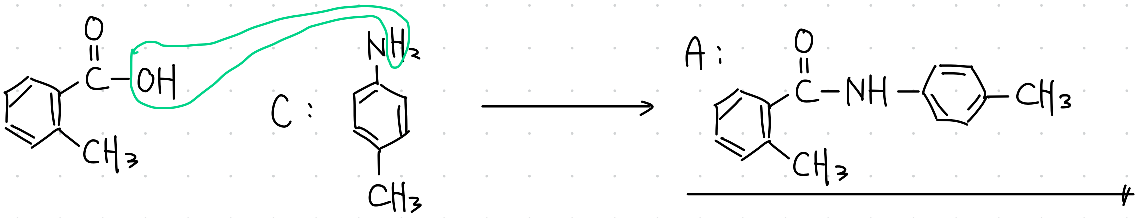


−NH₂, −CH₃ あり

$$\left(\begin{array}{c} -\mathrm{C}(=\mathrm{O})-\mathrm{NH}- \\ \downarrow \mathrm{H^+} \\ -\mathrm{C}(=\mathrm{O})-\mathrm{OH} \\ + \\ -\mathrm{NH_3^+Cl^-} \end{array}\right)$$

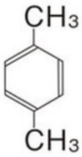
アミドA	カルボン酸	アミン
C_{15}	C_8以上	C_7以上
	↓	↓
	C_8	C_7

∴ B : C : A :

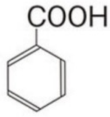


4 - $\boxed{9}$

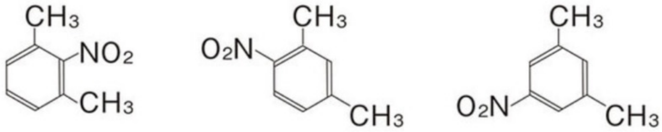
問 1　化合物 D：



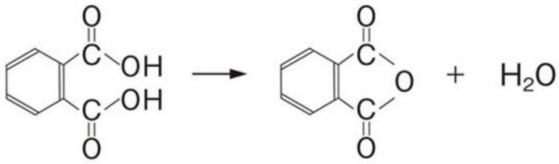
問 2　化合物 E：



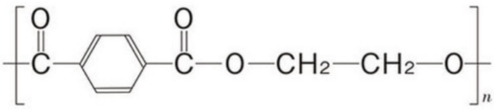
問 3



問 4



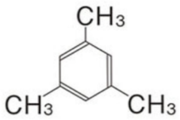
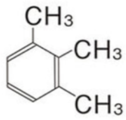
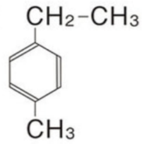
問 5

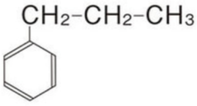
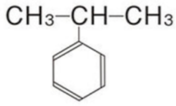
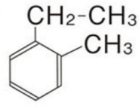


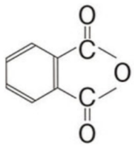
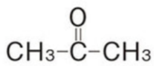
4 – $\boxed{10}$

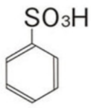
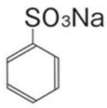
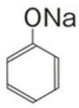
問 1（ア）　エチレングリコール（1, 2 エタンジオール）　　（イ）　酸素

（ウ）　ヨウ素

問 2　A 　B 　C 

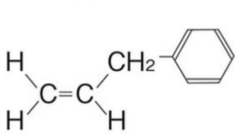
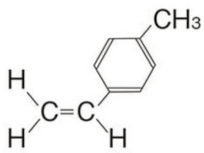
D 　E 　F 

K 　M 

問 3　N 　O 　P 

4 – $\boxed{11}$

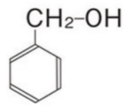
問 1　2 個　　**問 2**　クメン

問 3A 　E 

4－$\boxed{12}$

問1 気体…水素　　官能基…ヒドロキシ基

問2 試薬…$FeCl_3$　　色…紫

問3 X


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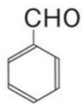

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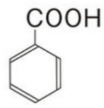
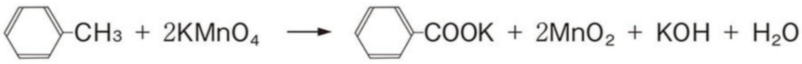
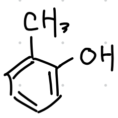


問4


4－$\boxed{13}$

問1　A：


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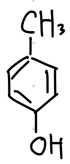

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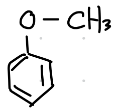

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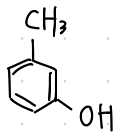

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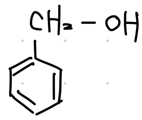
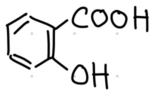
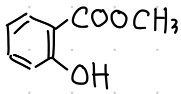


問2　NaOH aq を加えると A,B,D は中和して均一な溶液になる（とける）。
（C，E は 2層に分離する）　これだけでは×。見ためやにおいでわかることをかく。

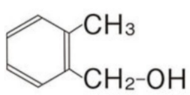
問3　ヒドロキシ基を持たないため，水素結合ができず，分子間力が弱いため。

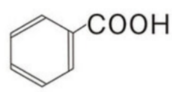
問4　金属 Na を加えたとき，C は H_2 を発生しない。

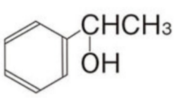
問5.　F：サリチル酸


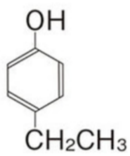
G：サリチル酸メチル


4－$\boxed{14}$

(1)　5　　(2)


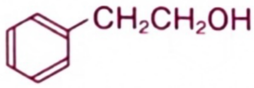
(3)　CHI_3　　(4)


(5)


(6)　9　　(7)


(8)　5　　(9)　1：2：2：2：3

4－ $\boxed{15}$

問1 A

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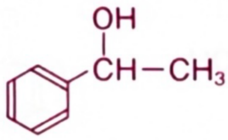
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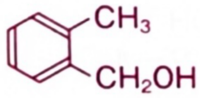
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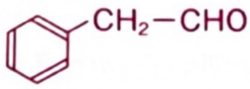
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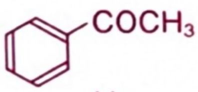
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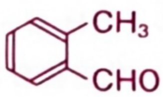
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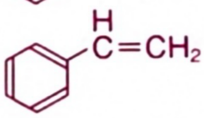
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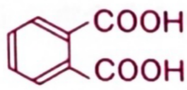
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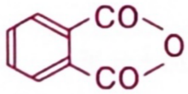
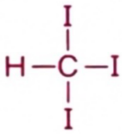
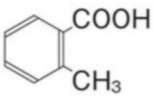


問2　B　　**問3**　ポリスチレン

問4　記号　B　構造式


4－ $\boxed{16}$

問1(1)
A

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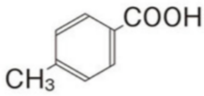
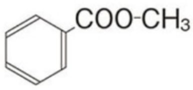


(2)　ナフタレン，o－キシレン

問2(1)
C

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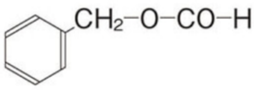
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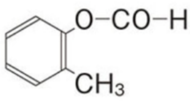
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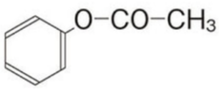
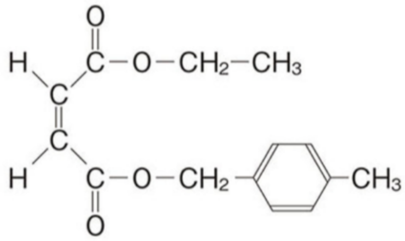


(2)　2,4,6－トリブロモフェノール

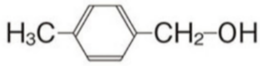
4 − $\boxed{17}$

問1 $C_{14}H_{16}O_4$

問2 A

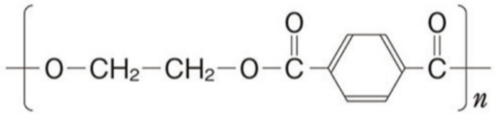


C



D H_3C-CH_2-OH

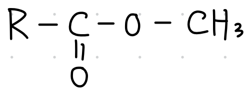
H



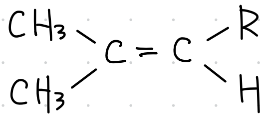
4 − $\boxed{18}$

問1 $C_{12}H_{14}O_2$

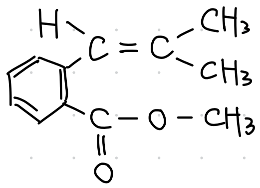
問2 (a)



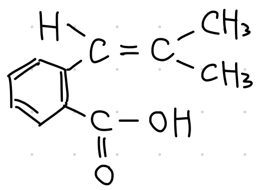
(b)



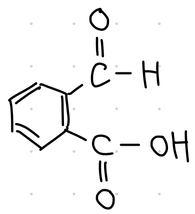
問3 A



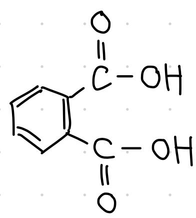
B



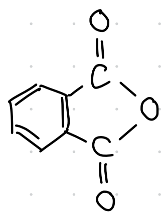
C



D



E



問1　$C_xH_yO_z$ とおくと,

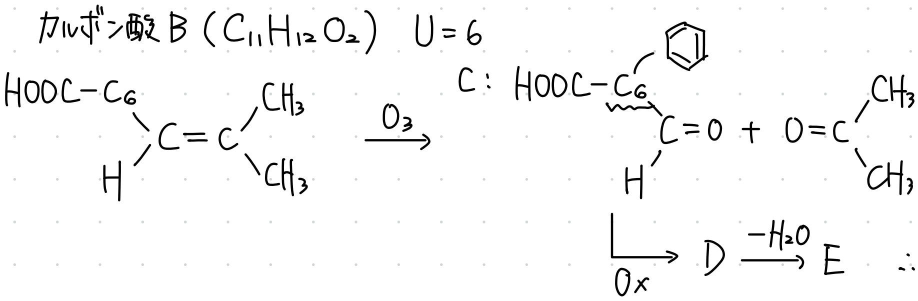
$$x = \frac{190\,g \times \frac{75.8}{100}}{12\ g/mol} = 12,\quad y = \frac{190\,g \times \frac{7.4}{100}}{1.0\ g/mol} = 14,\quad z = \frac{190\,g \times \frac{16.8}{100}}{16\ g/mol} = 2$$

∴ $C_{12}H_{14}O_2$　$U = 6 =$  $\times 1 + C{=}C \times 1 + -\overset{O}{\overset{\|}{C}}-O-$

$$A \xrightarrow{\text{加水分解}} \begin{cases} \text{カルボン酸}B\ (C_{11}H_{12}O_2) \\ \text{メタノール} \end{cases}$$

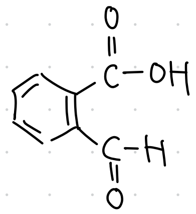
$\uparrow$ $A + H_2O - CH_4O$

カルボン酸B ($C_{11}H_{12}O_2$)　$U=6$

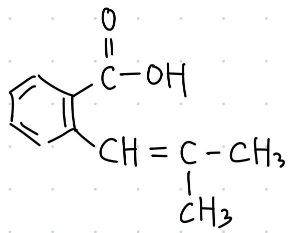


∴ Cはオルト体

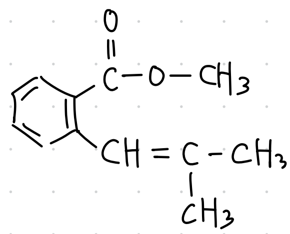
C:



B:



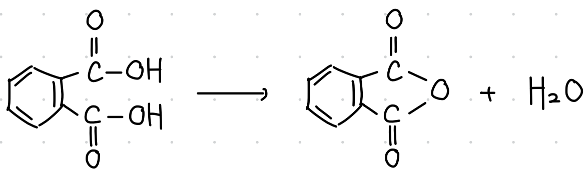
A:



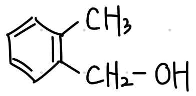
4 - $\boxed{19}$

問1　あ　C_5H_6O　　い　164　　う　$C_{10}H_{12}O_2$

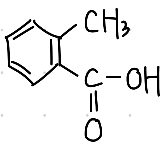
問2



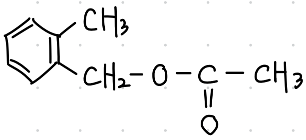
問3　C



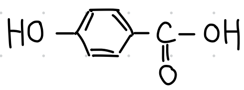
D



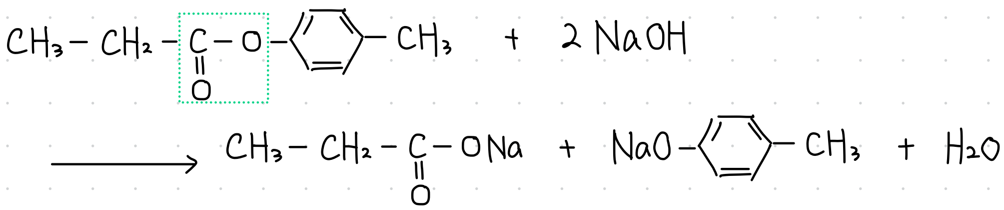
問4.



問5



問6



問1　$\Delta t_f = K_f m$　　$2.56 = 5.12 \times \dfrac{\dfrac{410\times10^{-3}\,g}{M\ g/mol}}{5\times10^{-3}\,kg}$　　∴ $M = 164$ (い)

$C_xH_yO_z \xrightarrow{O_2} x\,CO_2 + \frac{y}{2}H_2O$ より

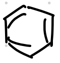
$C_xH_yO_z : CO_2 = 1 : x$ だから,　$\dfrac{8.2\,g}{164\,g/mol} \times x = \dfrac{22\,g}{44\,g/mol}$　∴ $x = 10$

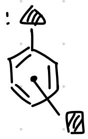
$C_xH_yO_z : H_2O = 1 : \frac{y}{2}$ だから,　$\dfrac{8.2\,g}{164\,g/mol} \times \dfrac{y}{2} = \dfrac{5.4\,g}{18\,g/mol}$　∴ $y = 12$

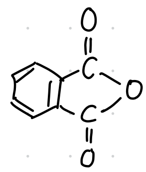
$M = 12x + y + 16z = 120 + 12 + 16z = 164$　∴ $z = 2$

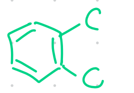
∴ $C_{10}H_{12}O_2$ (う)　　C_5H_6O (あ)

↓

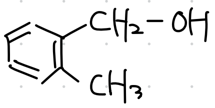
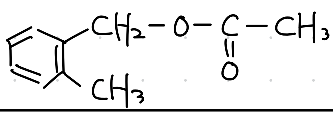
$U = 5 =$  $\times 4 + -C(=O)-O-$

芳香族エステルA $\xrightarrow{けん化}$ 工　C: 

- アルコール (~~フェノール類~~) : $-C-OH$
- $C \xrightarrow[弱く]{Ox}$ 酸性D ($C_8H_8O_2$) カルボン酸　C_8
- $C \xrightarrow{KMnO_4} E \xrightarrow{-H_2O} F$ ($C_8H_4O_3$) : 　オルト体



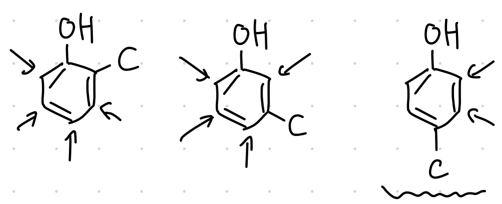
→ 水　C′ : カルボン酸のNa塩
$C_{10} - C_8 = C_2$

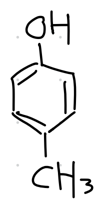
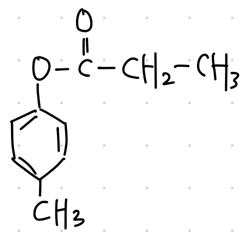
∴ C: 　　C′ : $CH_3-C(=O)-OH$　　∴ A: 

芳香族エステルB $\xrightarrow{けん化}$ 水 { フェノール類のNa塩 (G) / カルボン酸のNa塩 (G′) }

フェノール類のエステル　工 がない

$\xrightarrow{H^+}$ { フェノール類 油状G (酸性, 108) $\xrightarrow{KMnO_4}$ カルボン酸・フェノール類 H (138) }

G: 

∴ G: (108) 　　B: 

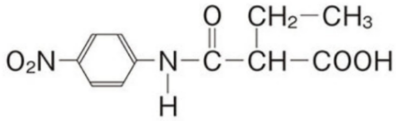
$G' = B + H_2O - G = C_{10}H_{12}O_2 + H_2O - C_7H_8O = C_3H_6O_2 = CH_3-CH_2-C(=O)-OH$

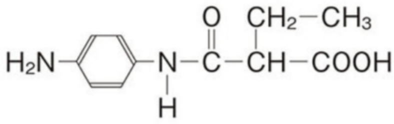
4－$\boxed{20}$

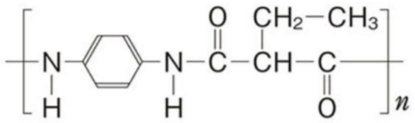
問1　C原子…52.4 %　　H原子…4.8 %　　N原子…11.1 %

問2　$RCOOH + NaHCO_3 \longrightarrow RCOONa + CO_2 + H_2O$

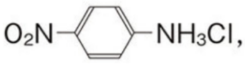
問3

A 　　B 

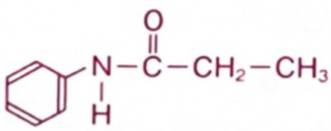
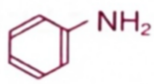
C　$CH_3-CH_2-CH(COOH)_2$　　D 

E 

問4　塩酸を用いることで化合物Aのアミド結合が加水分解されてしまうため。

生成物　，$CH_3-CH_2-CH(COOH)_2$

4－$\boxed{21}$

問1　A 　　B 

C　$CH_3-CH_2-C(=O)-OH$　　D　$CH_3-CH_2-CH_2-OH$

問2 ，31 mg

問3　F　$CH_3-CH_2-O-CH_3$，エチルメチルエーテル

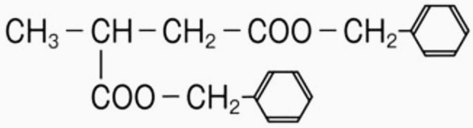
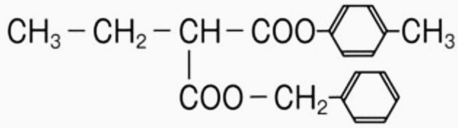
G　$CH_3-C(=O)-CH_3$，アセトン

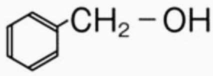
問4　F，E，D

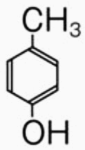
問5　H　$CH_3-CH(CH_3)-O-CH(CH_3)-CH_3$　　I　$CH_2=CH-CH_3$

4 − $\boxed{22}$

問1　$C_{19}H_{20}O_4$

問2　A：　B：

C：　D：$CH_3-CH(COOH)-CH_2-COOH$

E：　F：$CH_3-CH_2-CH(COOH)-COOH$　G：